

Switching to a 12V solenoid valve is the correct engineering decision when connecting directly to a pressurized water line. Submersible pumps cannot handle pressurized mains, and they lack the mechanical strength to hold back city water pressure when turned off.

A solenoid valve uses an electromagnetic coil to lift a physical plunger, allowing water to flow.

When power is cut, a heavy spring slams the plunger shut, sealing the pipe completely.

Here is the complete blueprint to safely bridge the 12V/5V voltage gap, wire the system, and protect your custom electronics from heavy monsoon rain, humidity, and intense UV sunlight.

Phase 1: The Components List (The 12V Upgrade)

You must redesign your power architecture. You will power the *entire* system from a single 12V adapter and step the voltage down internally for the microcontroller.

The Plumbing & Power:

- **12V Solenoid Valve (Normally Closed):** Look for a 1/2-inch brass or heavy-duty plastic solenoid. "Normally Closed" (N.C.) means if the power goes out, the valve stays shut, preventing a flood.
- **12V 2A DC Power Supply:** A standard router-style power brick.
- **LM2596 Buck Converter:** To step the 12V down to 5V for the ESP32.

The Control Logic:

- **ESP32-C3 SuperMini**
- **5V 1-Channel Relay Module (with Optocoupler):** This acts as the bridge. The ESP32 tells the relay to switch, and the relay safely passes the heavy 12V current to the solenoid.
- **1N4007 Diode (CRITICAL):** Solenoids are giant inductors. When turned off, the magnetic field collapses and shoots a massive reverse-voltage spike (inductive kickback) back up the wire. This \$1 Taka diode absorbs that spike. Without it, your ESP32 and relay will fry within a week.
- **Capacitive Soil Moisture Sensor (v1.2)**

Phase 2: Bridging the Gap (Wiring & Schematic)

You will split the power into two distinct "rails" inside your enclosure: a High-Power 12V Rail and a Low-Power 5V Rail.

1. Create the 5V Logic Rail:

- Connect the **12V Power Supply** to the IN+ and IN- of the **LM2596 Buck Converter**.
- *Mandatory Step:* Before connecting the ESP32, use a multimeter to tune the LM2596's brass screw until the output is exactly **5.0V**.
- Connect the LM2596 OUT+ to the ESP32-C3 **5V** pin and the Relay Module **VCC**.
- Connect the LM2596 OUT- to the ESP32-C3 **GND** and the Relay Module **GND**.

2. Wire the 12V Solenoid Rail:

- Split the incoming **12V positive wire** and connect it to the Relay's **COM** (Common) terminal.
- Connect the Relay's **N.O.** (Normally Open) terminal to the **Solenoid Positive (+)** wire.
- Connect the **Solenoid Negative (-)** wire straight back to the main **12V Ground**.

3. Install the Flyback Diode (1N4007):

- Install the diode directly across the two solenoid wires.
- The side of the diode with the **silver stripe** must point toward the **Positive (+)** wire coming from the relay. The blank side points to Ground.

4. Data Wiring:

- Connect ESP32-C3 **GPIO 10** to the Relay Module **IN** pin.
- Connect the Soil Moisture Sensor to **3V3**, **GND**, and **GPIO 0** (Analog).

Phase 3: The Weatherproofing Protocol

Electronics on a balcony face three enemies: liquid water, condensing humidity, and UV radiation. You cannot put this in a cardboard box or a thin 3D-printed PLA case.

1. The Enclosure:

- Buy an **IP65 or IP67 ABS Plastic Waterproof Junction Box**. Do not use PVC; ABS plastic is highly resistant to the UV degradation caused by direct sunlight.

2. Sealing the Exits (Cable Glands):

- You cannot just drill holes for your wires and fill them with hot glue. Hot glue melts in the summer sun and peels away, letting water in.
- Buy **PG7 or PG9 Nylon Cable Glands**. You drill a clean hole in the box, screw in the gland, pass your wires through, and tighten the rubber gasket down over the wire for a 100% watertight seal.

3. The "Drip Loop" Trick:

- Water flows down wires via surface tension. If a wire goes straight down into your box, rainwater will ride the wire directly into the gland.
- Always tie a **Drip Loop**. Let the wires droop lower than the junction box before looping back up to enter the cable gland. Gravity forces the water to drip off the bottom of the loop instead of entering the box.

4. Internal Humidity Protection (Conformal Coating):

- Even in a sealed box, high ambient humidity will eventually cause condensation on the copper traces.
- Buy a bottle of **Conformal Coating** (or simply use clear acrylic nail polish). Paint it over the ESP32-C3 board and the exposed solder joints of your moisture sensor (do NOT paint the actual sensor prongs going into the dirt). This creates a permanent, waterproof plastic shell over the silicon.

Phase 4: The Firmware & Home Assistant Config

Because you are using a relay just like the water pump setup, the C++ code remains almost exactly the same as the previous iteration.

ESP32-C3 Code Adjustment:

```
// Ensure the relay pin is defined and defaults to OFF
#define RELAY_PIN 10

void setup() {
    pinMode(RELAY_PIN, OUTPUT);
    digitalWrite(RELAY_PIN, LOW); // LOW keeps the N.O. relay open
    (Solenoid Closed)
    // ... rest of Wi-Fi and MQTT setup
}
```

Home Assistant configuration.yaml: Update your Home Assistant configuration to reflect the new hardware architecture.

```
mqtt:
  sensor:
    - name: "Balcony Soil Moisture"
      state_topic: "home/balcony/moisture"
      unit_of_measurement: "%"
      device_class: humidity

  switch:
    - name: "Balcony Solenoid Valve"
      state_topic: "home/balcony/valve/state"
      command_topic: "home/balcony/valve/set"
      payload_on: "ON"
      payload_off: "OFF"
      icon: mdi:water-pump
```

Once this is flashed and sealed in the IP67 box, you will have a fully autonomous, weather-proof, direct-line irrigation system that you can monitor and trigger from anywhere in the world via your Raspberry Pi server.